

USTHB University
Faculty of Computer Science
AI&DS Department

DevOps & Cloud Computing

01- Virtualization







Engineers (5th Year) | AMEUR IMANE | 2026/2027

DevOps and Cloud Computing: Engineers (5th Year) : Software Engineering

5- Semestre 9 :

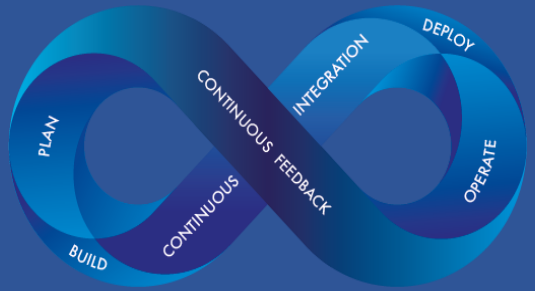
Unité d'Enseignement	VHS	V.H hebdomadaire				Coeff.	Crédits	Mode d'enseignement		Mode d'évaluation	
	14-16 sem	C	TD	TP	Autres			A distance	En présentiel	Continu	Examen
UE fondamentales											
UEF1(O/P)											
Méthodes formelles pour GL	42h	1h30	1h30/15J	1h30/15J		3	4		X	40%	60%
Développement de logiciels embarqués	42h	1h30		1h30		3	4		X	40%	60%
UEF2(O/P)											
Conception de jeux vidéo : Théorie et Pratique	42h	1h30		1h30		3	4		X	40%	60%
Internet of Things (IoT) : Concepts et développement	63h	1h30	1h30	1h30		3	5		X	40%	60%
UE méthodologie											
UEM1 (O/P)											
DevOPs & Cloud Computing	84h	3h00		3h00		3	5		X	40%	60%
Sécurité Logicielle	42h	1h30		1h30		2	3		X	40%	60%
Développement mobile	42h	1h30		1h30		2	3			40%	60%
UE transversales											
UET1(O/P)											
Aspects juridiques	21h	1h30				1	2		X		100%
Total Semestre 9	**378h**	**13h30**	**2h15**	**11h15**		**20**	**30**				

- Volume hebdomadaire = 27H00 / semaine.
- 8 Matières par semestre.



COURSE OUTLINE

What you'll learn in this course



01 **Principles of virtualization**
What virtualization is and why it matters

02 **Objectives and benefits**
Why organizations virtualize

03 **History**
From mainframes to the cloud

04 **Definitions, concepts and techniques**
VM, hypervisor, virtualization types

05 **OS virtualization architectures**
Type 1, Type 2, OS-level

06 **Virtualization in different domains**
Server, desktop, network, storage and application

07 **Performance and efficiency**
Overhead and optimization techniques

1. Principles of virtualization



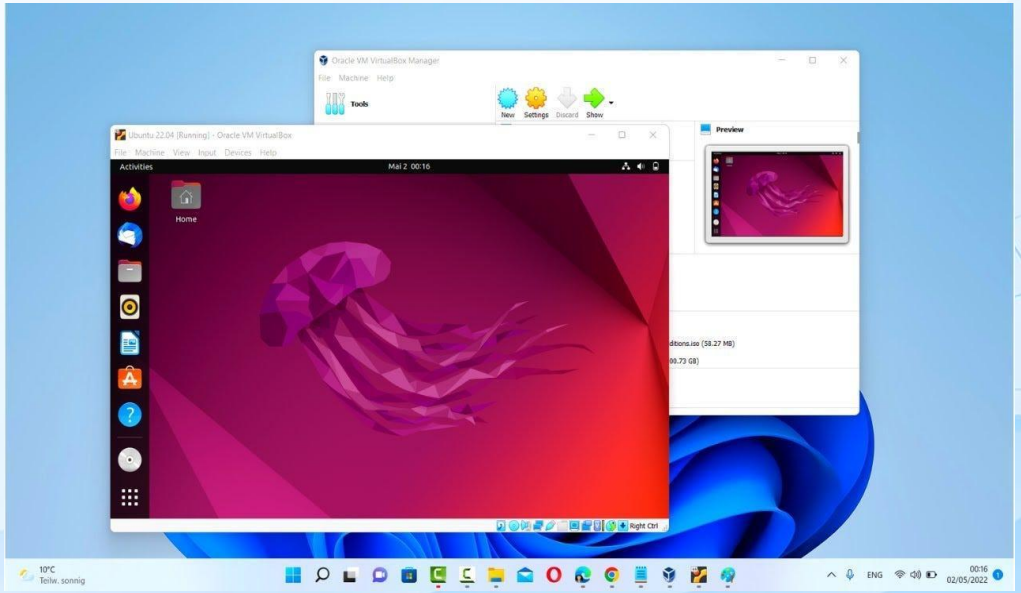
What virtualization is, and why it changes everything

Introduction

Virtualization: The University Computer Lab

"Do we need to buy a second physical computer for every student who wants to use two operating systems?"

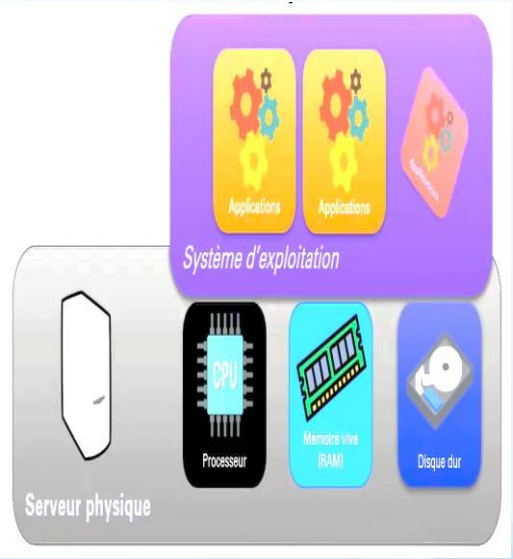
Instead of buying another physical computer, we can use software to create a virtual computer inside the existing computer.

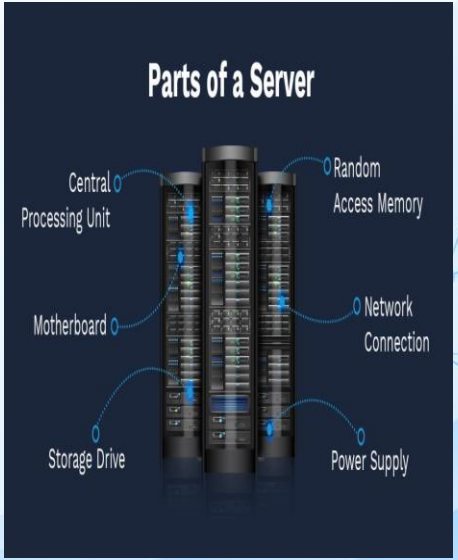


Introduction

The physical server is essentially composed of:

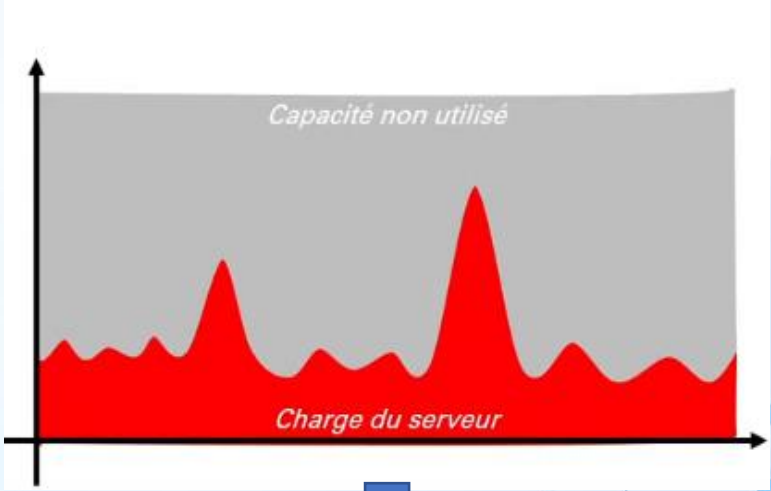
- **a CPU**
- **RAM (random access memory)**
- **a hard disk**
- **an operating system**





Introduction

By observing the typical use of a server, we see that it is often largely underutilized, with much of its computing power and memory left unused; this is why virtualization was invented.



One Application per Server: Historically, businesses would dedicate **one application** to **one physical server**.

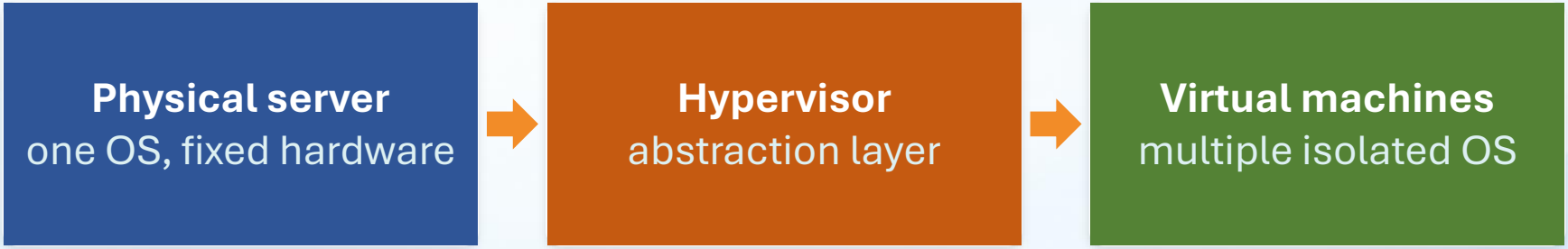
Virtualization

What is virtualization?

Virtualization is the process of creating a virtual version of a physical resource; such as a server, storage device, or network; using software to improve efficiency and resource utilization.

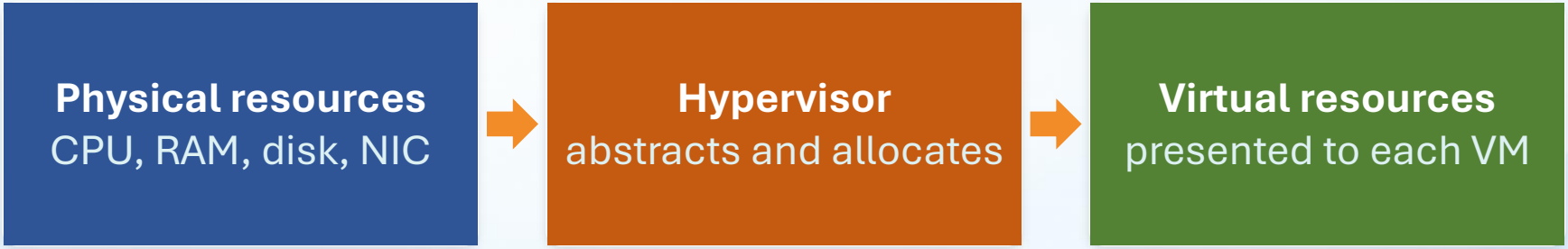
- So, creating a **virtual (software-based) version** of a computing resource rather than **a real physical one.**

What is virtualization?



- Virtualization creates a software-based (virtual) version of hardware, an OS, storage or a network
- One physical machine can run several independent virtual machines (VMs)
- Each VM behaves as if it had its own dedicated hardware

Core principle: the abstraction layer



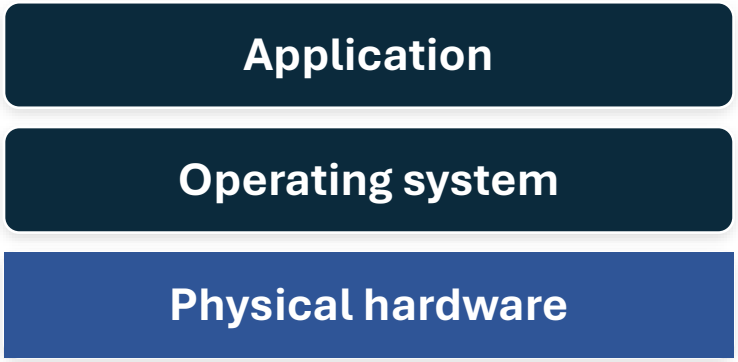
- The hypervisor intercepts hardware calls and shares physical resources among VMs
- Each VM believes it owns the hardware exclusively
- This abstraction is the foundation of cloud computing

Traditional vs virtualized architecture

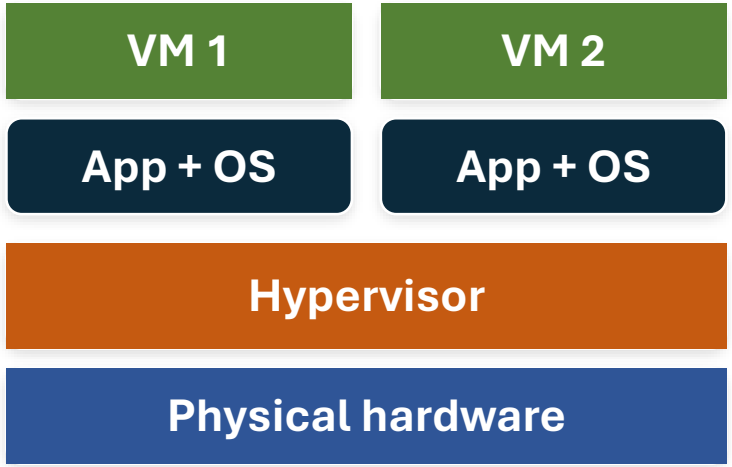




Traditional



Virtualized



One physical machine now runs several independent operating systems

2. Objectives and benefits



Why organizations virtualize

Objectives and benefits of virtualization

1. **Server consolidation**
   Run many VMs on fewer physical machines
2. **Cost reduction**
   Less hardware, power, cooling, space
3. **Flexibility**
   Create, resize or move a VM in minutes
4. **Isolation**
   A crash in one VM does not affect others
5. **Disaster recovery**
   VMs are easy to back up and restore
6. **Testing and development**
   Safe, disposable environments

Server consolidation: before and after

Before: 6 physical servers, low utilization

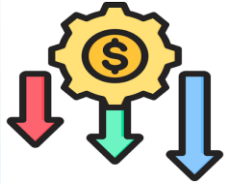
Server 1 ~15% used	Server 2 ~15% used	Server 3 ~15% used	Server 4 ~15% used	Server 5 ~15% used	Server 6 ~15% used

After: 2 physical hosts running the same workloads as VMs

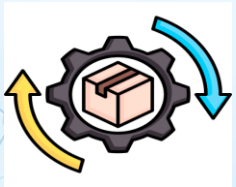
Host 1 — Hypervisor 3 VMs, ~70% used	Host 2 — Hypervisor 3 VMs, ~70% used

Same workloads, far fewer physical machines, much better utilization

Why virtualize?



- **Cost Reduction :** Instead of buying 10 physical servers for 10 tasks, you buy only 1 or 2 and virtualize them. This reduces hardware purchases, energy consumption, and required space.



- **Better Utilization :** Most physical servers are underutilized (running at 5–15% of their capacity). Virtualization allows much more of this unused power to be harnessed.



- **Flexibility :** Virtual Machines can be easily created, saved (via snapshots), and moved from one physical server to another, greatly simplifying maintenance and disaster recovery.

Quantifying the benefits



Benefit	Typical impact
Hardware count	Reduced by a factor of 3 to 10
Provisioning time	From weeks to minutes
Energy and space	Significantly reduced
Recovery time	Minutes, via VM snapshot or migration

3. History



From mainframes to the cloud

Timeline of virtualization



1960s

IBM CP-40/CP-67: first virtual machines on mainframes

1990s

x86 virtualization research; VMware founded (1998)

2000s

Xen and KVM: open-source hypervisors for Linux

2006

Amazon EC2: virtualization becomes a commercial cloud service

2013+

Containers (Docker) popularize lightweight virtualization

Why virtualization exploded in the 2000s

1. **Hardware-assisted virtualization**
   Intel VT-x and AMD-V (2005-2006) made x86 virtualization fast and reliable



2. **Growing server sprawl**
   One application per server was wasteful and hard to manage



3. **Rise of the internet**
   More services needed to be hosted flexibly



History of Virtualization



1960: The Beginnings

IBM developed CP/CMS (Control Program/Cambridge Monitor System), an operating system that allowed the creation of virtual machines (VMs). **This was the first approach to virtual machines.**

1970: Hypervisors

IBM VM/370 introduced hypervisors; **a software layer to run multiple systems on a single hardware.**

1990: x86 Virtualization

VMware (1998) launched the first x86 virtualization product, "VMware Virtual Platform," **marking the rise of virtualization on x86 architecture.**

History of Virtualization



2000: Enterprise Virtualization Boom

VMware ESX Server (2001) enabled server virtualization for enterprises.

Emerging competitors included Microsoft Hyper-V and XenSource with Xen.

2010: Containers and Lightweight Virtualization

Docker (2010) introduced containers, **offering faster deployment and greater efficiency compared to traditional VMs.**

2015: Continued Innovation

Kubernetes (version 1.0 released on July 21, 2015) enabled container orchestration.

There was growing adoption of network and storage virtualization.

4. Definitions and concepts



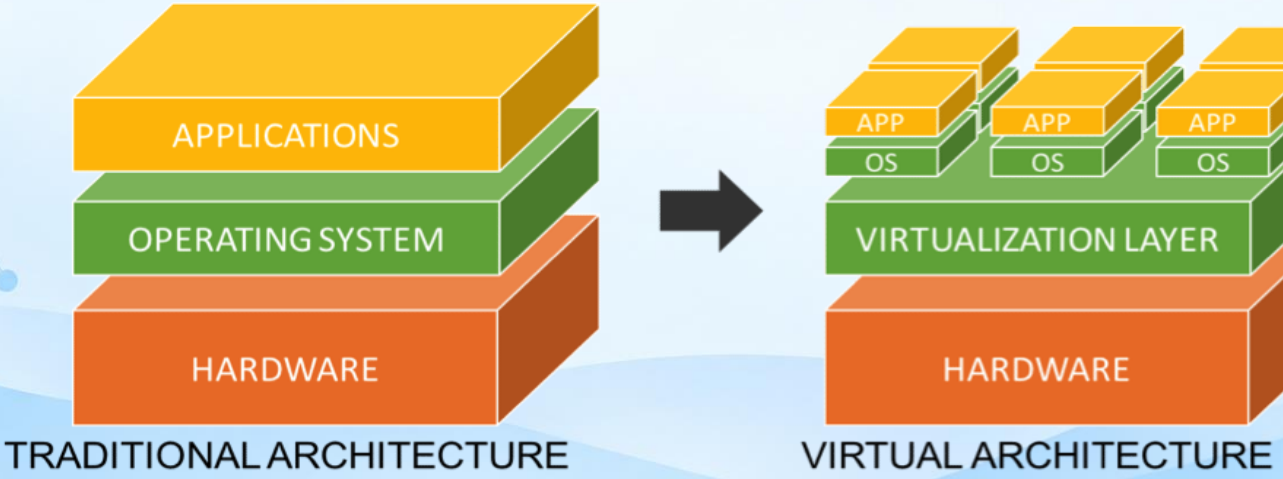
VM, hypervisor and virtualization techniques

Key vocabulary

Term	Definition
Host	The physical machine running the hypervisor
Guest / VM	A virtual machine running its own OS
Hypervisor	Software layer that creates and manages VMs
Virtual disk	A file that represents a VM's hard drive
Snapshot	A saved state of a VM at a point in time

Definition

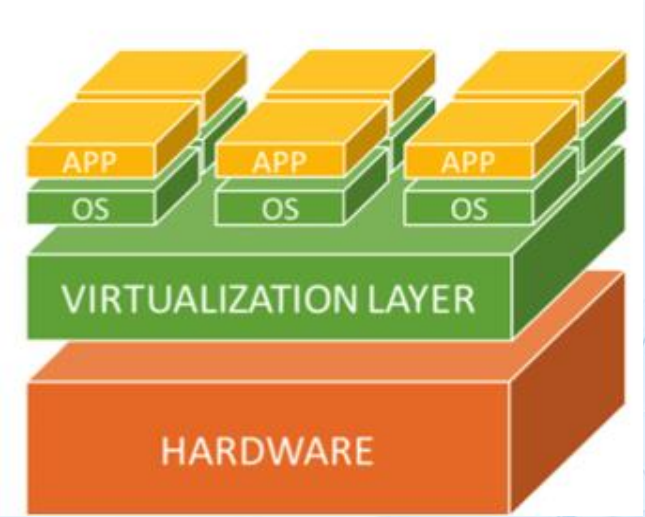
Virtualization consists of creating **multiple Virtual Machines (VMs)** from **a single physical machine** through an abstraction layer close to the hardware. This ***abstraction layer*** is provided by the **hypervisor**.



Hypervisor

The **hypervisor** sits between the **physical hardware** and the **virtual operating systems (the VMs).**

It hides the **complex details** of the **hardware** from the **VMs**, allowing **any operating system** to **run** on **any hardware** (hardware independence).



Type 1 hypervisor: bare-metal

Hardware
CPU, RAM, disk

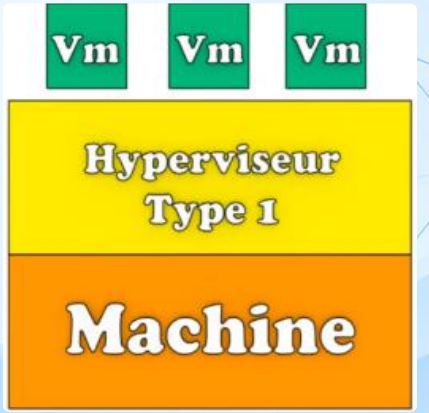
→

Hypervisor
runs directly on hardware

→

VM 1 / VM 2 / VM 3
each with its own OS

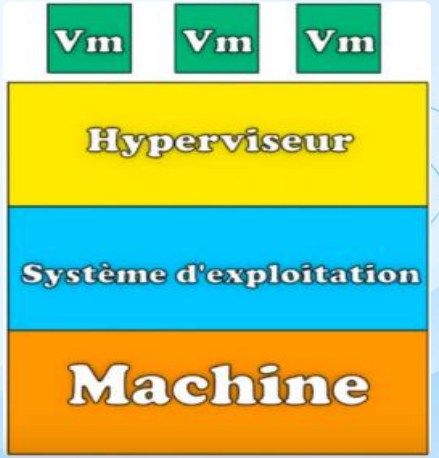
- No host operating system underneath the hypervisor
- Best performance, used in data centers and the cloud
- Examples: VMware ESXi, Microsoft Hyper-V, Xen, KVM



Type 2 hypervisor: hosted



- The hypervisor runs as a normal application inside a host OS
- Simple to install, ideal for desktops and testing
- Examples: VMware Workstation, VirtualBox, Parallels



Type 1 vs Type 2: comparison

Criterion	Type 1 (bare-metal)	Type 2 (hosted)
Runs on	Hardware directly	A host operating system
Performance	Higher	Lower (extra OS layer)
Typical use	Data centers, cloud	Desktops, development, testing
Examples	ESXi, Hyper-V, KVM, Xen	VirtualBox, VMware Workstation

VMware and Hyper-V

Hypervisors like VMware and Hyper-V are virtualization technologies that allow multiple virtual machines (VMs) to run on a single physical server.
Here is a brief overview of these two popular hypervisors:

a. VMware is one of the leading providers of virtualization solutions, offering products such as VMware vSphere and VMware Workstation.

- **VMware vSphere** is a virtualization platform for data centers that enables centralized management of IT resources, virtual machines, and applications.
- **VMware Workstation** is designed for individual users and developers, allowing them to create and manage VMs on a desktop computer.
- VMware also offers other products, such as VMware ESXi (a free hypervisor) and VMware Fusion (for Mac users).

VMware and Hyper-V

b. Hyper-V is a hypervisor developed by Microsoft and has been integrated into Windows Server since Windows Server 2008.

It is also available as a standalone product called “**Hyper-V Server**” and is included in Windows 10 Pro and Enterprise editions for use on desktops.

What is virtualization?

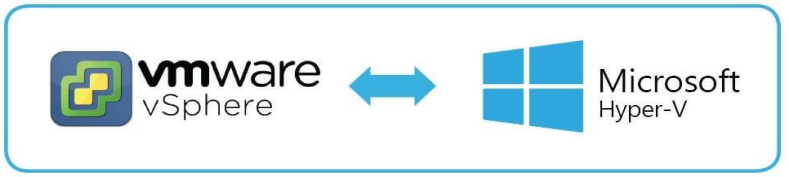
Installing Hyper-V on Windows 10

https://learn.microsoft.com/fr-fr/virtualization/hyper-v-on-windows/quick-start/enable-hyper-v

Creating a Virtual Machine with Hyper-V on Windows 10

https://learn.microsoft.com/fr-fr/virtualization/hyper-v-on-windows/quick-start/quick-create-virtual-machine

Hyper-V vs. VMware



Hypervisor	Advantages	Disadvantages
Hyper-V	Free, easy to use, good performance	Fewer features than VMware, less well supported than VMware
VMware	More features, better support	More expensive than Hyper-V, more complex to use

The choice between VMware and Hyper-V often depends on the organization's specific needs, hardware, and user preference.

5. OS virtualization architectures



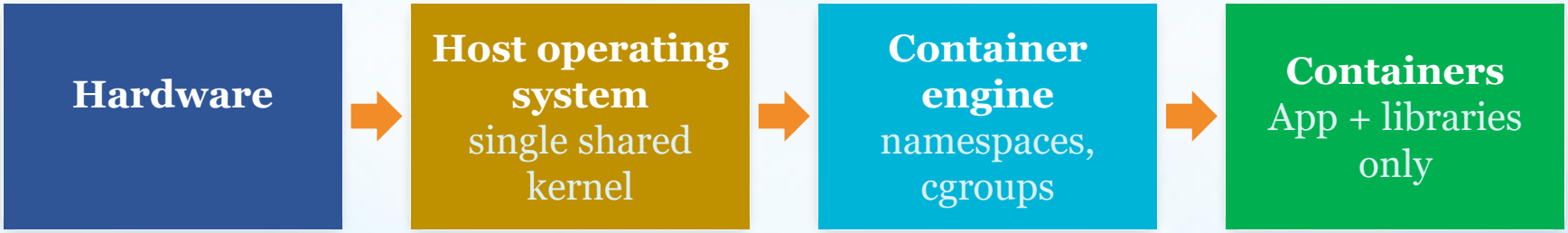
Comparing the three architectures visually

Architecture: full virtualization (Type 1)



- Each VM runs a complete, independent operating system
- Strong isolation, but each OS consumes its own memory and CPU cycles

Architecture: Virtualization based containers



- No guest OS: containers share the host kernel
- Much lighter and faster to start than VMs
- Trade-off: less isolation than a full VM

6. Virtualization in different domains



Server, desktop, network, storage, application

Domains of virtualization



VIRTUALISATION
DE SERVEURS



VIRTUALISATION
DES SYSTÈMES D'EXPLOITATION



VIRTUALISATION
DE STOCKAGE



VIRTUALISATION
DES POSTES DE TRAVAIL

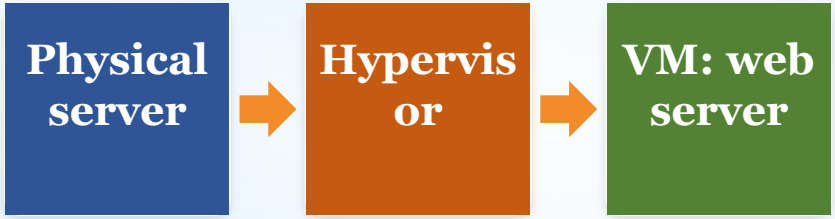


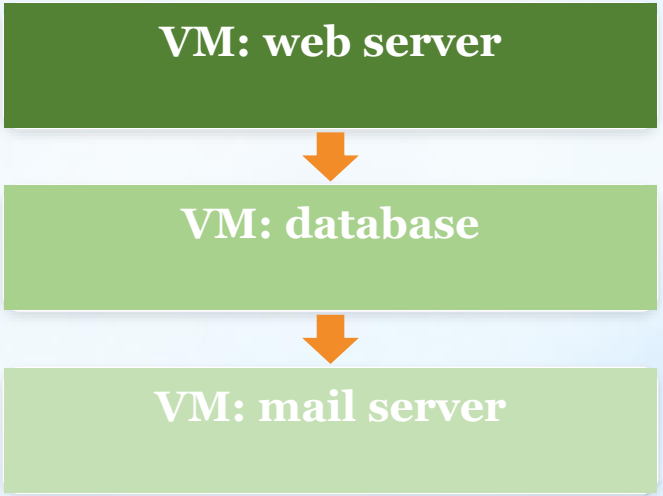
VIRTUALISATION
D'APPLICATIONS



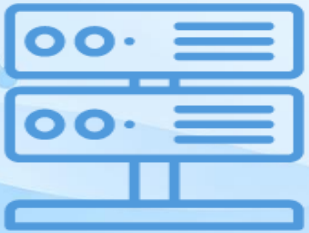
VIRTUALISATION
DES RÉSEAUX

Server virtualization





- The most common form: consolidating multiple server roles on shared hardware



allows **multiple virtual servers** to run on your **physical server**, multiplying operating systems and enabling full utilization of your hardware.

Desktop virtualization

- Users access a full desktop OS running remotely
- Benefits: centralized management, easier security, access from anywhere
- Examples: Citrix Virtual Apps, VMware Horizon



allows different company employees **to connect,** via a secure session**, to a work environment** where their data as well as their software are centralized.

University of Michigan: The university adopted the **Citrix Virtual Apps and Desktops** solution to provide virtual desktops to its students. This allows students to access specialized software (such as **AutoCAD or MATLAB**) from any device, even if they do not have computers powerful enough to run these applications locally.

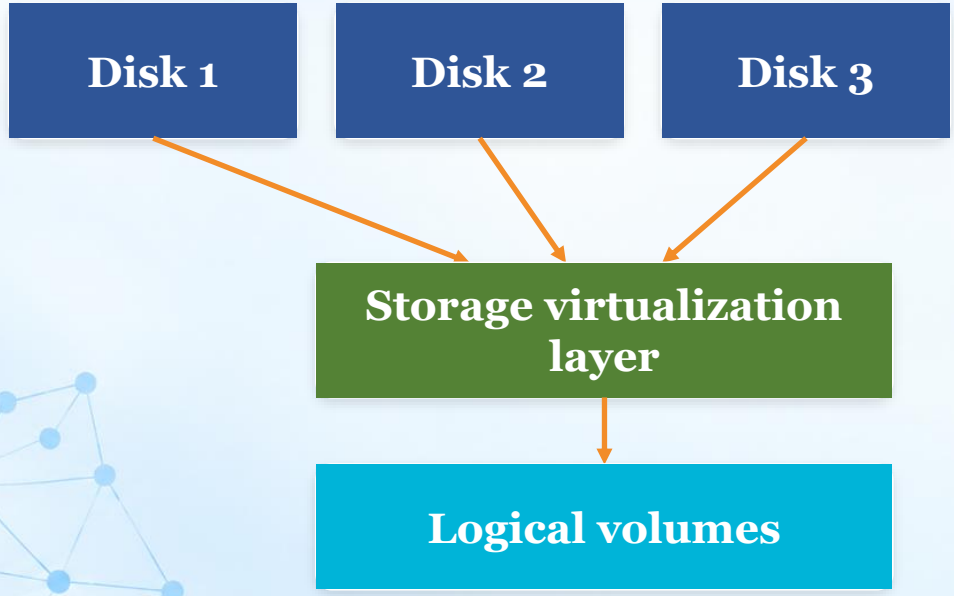
Network virtualization

Traditional	Virtualized Network
Dedicated hardware router	Software router (virtual appliance)
Dedicated hardware firewall	Software firewall (virtual appliance)
Dedicated hardware load balancer	Software load balancer



allows the virtualization of **network services** such as routers, firewalls, load balancers, and more.

Storage virtualization



is a technology that allows the creation of virtual storage volumes from multiple physical disks.



Netflix uses **storage virtualization** in its cloud infrastructure to manage the massive amount of video data and deliver it to users.

Application virtualization

- An application runs in an isolated environment, without being fully installed on the OS
- Avoids conflicts between application versions and dependencies
- Examples: Microsoft App-V, containerized desktop apps

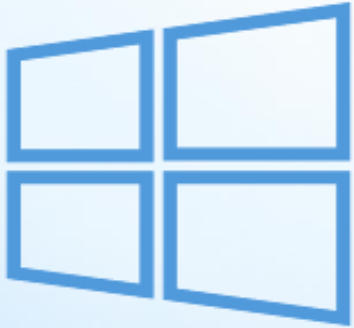


is a technique that allows applications to run in isolation, independently of the underlying operating system environment.
Application virtualization can be implemented using **containers**.

Imagine a company that uses multiple versions of Microsoft Office. Traditionally, this could cause file conflicts between versions (for example, different versions of Excel or Outlook installed on the same machine). Thanks to application virtualization with **Microsoft App-V**, the company can virtualize **different versions of Microsoft Office** and run them simultaneously without conflict, with each version isolated in its own virtual environment..

Operating system virtualization

It is possible that you may need to use multiple different operating systems; such as Windows, Linux, Mac OS X, etc. on the same computer.



allows **multiple isolated instances** of an **operating system** to run on **a single physical machine.**

7. Performance and efficiency



Overhead, and how to reduce it

Where does virtualization overhead come from?

1

CPU scheduling
The hypervisor shares CPU time among VMs

2

I/O virtualization
Disk and network access pass through an extra layer

3

Memory management
Extra translation between virtual and physical memory

4

Context switching
Switching between VMs costs CPU cycles

Hardware-assisted virtualization

Without CPU extensions
software-only translation, higher overhead

With Intel VT-x / AMD-V
hardware executes guest instructions directly

- Modern CPUs include dedicated virtualization instructions
- This dramatically reduces the performance gap between virtual and physical machines

Techniques to improve performance

1. **Paravirtualized drivers**
   Guest OS uses optimized virtual I/O drivers

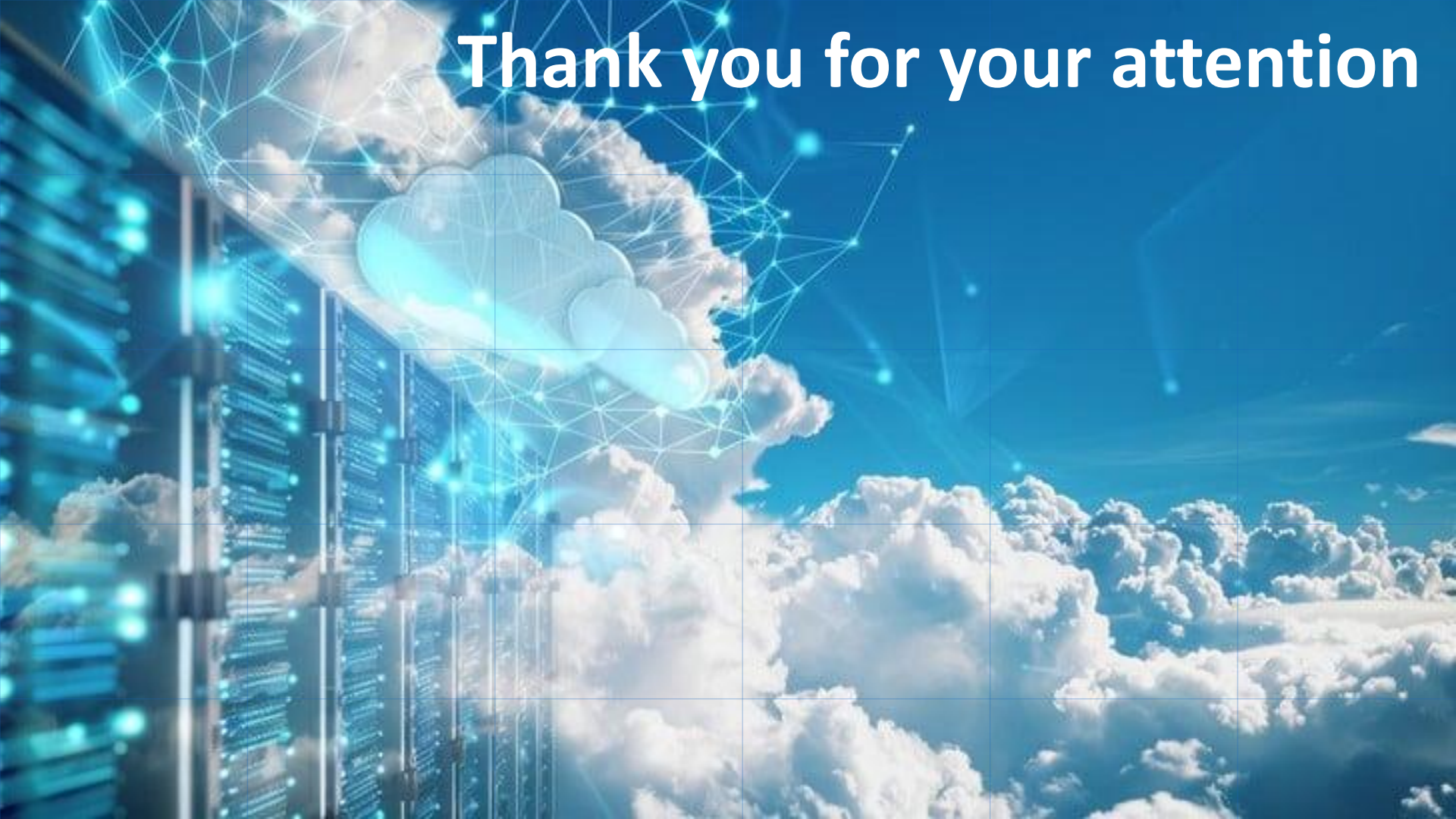
2. **Memory ballooning**
   Reclaims unused RAM from idle VMs

3. **CPU pinning**
   Dedicates specific CPU cores to a VM

4. **SR-IOV**
   Lets a VM access network hardware directly

Choosing the right approach: a checklist

Question	If yes, consider…
Need to run different operating systems?	Virtual machines
Need maximum density and fast startup?	Containers
Need strongest isolation between tenants?	Virtual machines
Running on a laptop for testing only?	Type 2 hypervisor
Running production workloads at scale?	Type 1 hypervisor or containers


Thank you for your attention